ICD-9-CM codes with no match in the target system

There are 9 ICD-9-CM codes with no ICD-10-CA match and 52 with no ICDA-8 match. The reference crosswalk lists them with no partner, so for these codes the correct output is no mapping at all.

The algorithm decides using two numbers. One is how often a pair of codes shows up on the same patient record. The other is the cosine similarity between the two code labels, which here comes from ClinicalBERT. Both are below, for each code on its own.

ICD-9-CM to ICD-10-CA, the 9 codes

Co-occurrence first, since that is the one we expected to be low. Partners is how many different ICD-10-CA codes each one turned up with, and the five columns after it describe the counts across those partners. Nothing below 7 is reported in the source data, which is why every minimum is 7.

Code	Label	Partners	Min	25th	Median	75th	Max	Most frequent partner
338	pain not elsewhere classified	827	7	18	49	181.5	13,431	E11, Type 2 diabetes mellitus
515	postinflammatory pulmonary fibrosis	306	7	13	24	53.8	5,194	J84, Other interstitial pulmonary diseases
239	neoplasms of unspecified nature	585	7	14	34	91	4,862	I10, Essential (primary) hypertension
327	organic sleep disorders	349	7	12	23	49	3,108	G47, Other sleep disorders
249	secondary diabetes mellitus	161	7	12	18	38	1,783	E11, Type 2 diabetes mellitus
209	neuroendocrine tumors	124	7	11.8	17	31.8	506	C78, Secondary malignant neoplasm of respiratory and digestive organs
175	malignant neoplasm of male breast	24	7	8	10.5	22.5	131	C79, Secondary malignant neoplasm of other and unspecified sites
339	other headache syndromes	58	7	9.2	15	17.8	54	E11, Type 2 diabetes mellitus
445	atheroembolism	2	7	9.2	11.5	13.8	16	I74, Arterial embolism and thrombosis

They are not low. Code 338, pain not elsewhere classified, turned up with 827 different ICD-10-CA codes, and with E11, type 2 diabetes mellitus, 13,431 times. Code 515 reaches 5,194 and code 239 reaches 4,862. For comparison I took the 345 ICD-9-CM codes that do have a match and found the middle value of their highest count, which is 1,288.5. 5 of these 9 are above it.

Two do behave the way we expected. Code 445, atheroembolism, has 2 partners and never gets past 16, and code 339, other headache syndromes, has 58 partners and a top count of 54.

The count seems to track how often a code gets written down, not whether it has a valid target. Pain, sleep disorders and unspecified neoplasms get recorded on all sorts of patients, so they sit next to whatever else the patient has.

Cosine similarity next. Each of the 9 is scored against all 2038 ICD-10-CA codes, so each row below is a summary of 2038 numbers.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
175	malignant neoplasm of male breast	0.825	0.641	0.796	0.826	0.849	0.955	C30, Malignant neoplasm of nasal cavity and middle ear
249	secondary diabetes mellitus	0.817	0.663	0.774	0.817	0.859	0.951	O24, Diabetes mellitus in pregnancy
239	neoplasms of unspecified nature	0.831	0.662	0.804	0.833	0.859	0.942	D15, Benign neoplasm of other and unspecified intrathoracic organs
515	postinflammatory pulmonary fibrosis	0.852	0.694	0.823	0.858	0.886	0.932	N72, Inflammatory disease of cervix uteri
209	neuroendocrine tumors	0.836	0.693	0.808	0.842	0.867	0.915	A74, Other diseases caused by chlamydiae
327	organic sleep disorders	0.770	0.541	0.725	0.773	0.823	0.915	P78, Other perinatal digestive system disorders
339	other headache syndromes	0.816	0.648	0.786	0.820	0.851	0.915	A69, Other spirochaetal infections
338	pain not elsewhere classified	0.805	0.685	0.785	0.807	0.827	0.893	R52, Pain, not elsewhere classified
445	atheroembolism	0.711	0.552	0.686	0.713	0.736	0.853	M66, Spontaneous rupture of synovium and tendon

The best score for these 9 runs from 0.853 to 0.955. For the 345 codes that do have a match it runs from 0.861 to 0.981. The two ranges sit almost on top of each other, so a high score on its own does not tell us whether a code has a real match.

The closest code is also wrong in 8 of the 9, and not narrowly wrong. Code 175 is malignant neoplasm of male breast and its closest ICD-10-CA code is C30, malignant neoplasm of nasal cavity and middle ear, at 0.955. Code 515 is postinflammatory pulmonary fibrosis and its closest is N72, inflammatory disease of cervix uteri. The model is picking up shared words rather than shared meaning.

The one worth a second look is code 338. Its closest target is R52, and both are labelled pain not elsewhere classified, but the crosswalk still records no match for it. I wanted to flag it in case that one is an error in the reference rather than in the algorithm.

ICD-9-CM to ICDA-8, the 52 codes

These behave completely differently. None of the 52 appear in the co-occurrence data at all. Not a low count, no row. A code is only listed there if it shares a record with at least one ICDA-8 code, and none of these ever did, so there is nothing to describe for that signal.

That works out in our favour, because it means the algorithm returns no match for all 52, which is the right answer. But it gets there for a reason that costs us elsewhere. It will not accept a pair without co-occurrence data, and 32 of the 302 ICD-9-CM codes that do have a valid ICDA-8 match are missing from that file too. It stays silent on those as well, and loses 39 mappings that are correct.

Similarity looks much the same as it did on the other crosswalk. Each code is scored against all 858 ICDA-8 codes. The best score for the 52 runs from 0.883 to 0.985, against 0.865 to 1.000 for the 302 that do have a match, so again there is no gap between the two groups. Rather than list all 52 I have put the four highest and the four lowest below, and I have the rest if you want them.

Code	Label	Average	Min	25th	Median	75th	Max	Closest target code
164	malignant neoplasm of thymus heart and mediastinum	0.813	0.584	0.776	0.810	0.846	0.985	142, malignant neoplasm of salivary gland
175	malignant neoplasm of male breast	0.816	0.578	0.782	0.816	0.848	0.984	174, malignant neoplasm of breast
237	neoplasm of uncertain behavior of endocrine glands and nervous system	0.821	0.594	0.785	0.822	0.855	0.984	238, neoplasm of unspecified nature of eye brain and other parts of nervous system
179	malignant neoplasm of uterus part unspecified	0.812	0.633	0.780	0.810	0.842	0.980	149, malignant neoplasm of pharynx unspecified
588	disorders resulting from impaired renal function	0.820	0.700	0.801	0.822	0.840	0.901	515, pneumoconiosis due to silica and silicates
312	disturbance of conduct not elsewhere classified	0.775	0.678	0.751	0.775	0.797	0.895	306, special symptoms not elsewhere classified
625	pain and other symptoms associated with female genital organs	0.762	0.540	0.727	0.768	0.807	0.893	781, other symptoms referable to nervous system and special senses
315	specific delays in development	0.777	0.671	0.752	0.776	0.800	0.883	066, late effects of viral encephalitis

Whether a threshold on either number would catch these

Not on its own. On ICD-10-CA a similarity cut-off high enough to clear all 9 would have to sit above 0.955, and that also removes 286 of the 345 codes that are correct. On ICDA-8 the cut-off would be 0.985 and it removes 175 of 302. Co-occurrence is no better. A count cut-off high enough to clear the 9 would have to sit above 13,431, which removes 293 of the 338 matched codes that are in the file. Both together do not help either, because on ICD-10-CA the 9 pass both tests comfortably, and on ICDA-8 the 52 are already being caught by the missing data rather than by any threshold.

What the algorithm is actually doing with these two numbers

While I was checking the above I found something I was not expecting, and I think it is a more useful answer to why the special cases are not being caught.

Neither threshold in the pipeline is an absolute one.

The similarity threshold is applied as the threshold multiplied by that code's own highest score, separately inside each ICD-9-CM column. The value we vary from 0.95 to 0.999 is a fraction of the code's own best score, not a score. So the top target of every code clears it however low the score actually is. The weakest code in the ICD-9-CM to ICD-10-CA set has a best score of 0.853 and it still produces a candidate. There is no value a cosine similarity can fall below and be rejected.

Co-occurrence works the same way. The rule keeps each code's top n partners ranked by frequency, with n varied from 3 to 30, and the count itself is never compared against anything. The smallest highest count in the file is 7 and it is kept on the same terms as the code with 238,548.

So the rule we describe as a cosine similarity above one value and a co-occurrence above another is not the rule the code is applying. Both are relative to the code being mapped. The only two things that make the algorithm stay quiet are the chapter filter removing every candidate, or, under the flag settings that require co-occurrence, the code being absent from the co-occurrence file. That second one is what is happening to all 52 on the ICDA-8 side, and it is why the right answer there is coming out for the wrong reason.

I think that is worth settling before we test the abstain scenarios, because those scenarios assume a floor that is not in the code yet. Putting a real absolute threshold in would be a change to the pipeline rather than a change to a parameter, and going by the numbers above it would cost a lot of correct mappings, so I have not made it.